Chapter Overview
Respiration in Plants

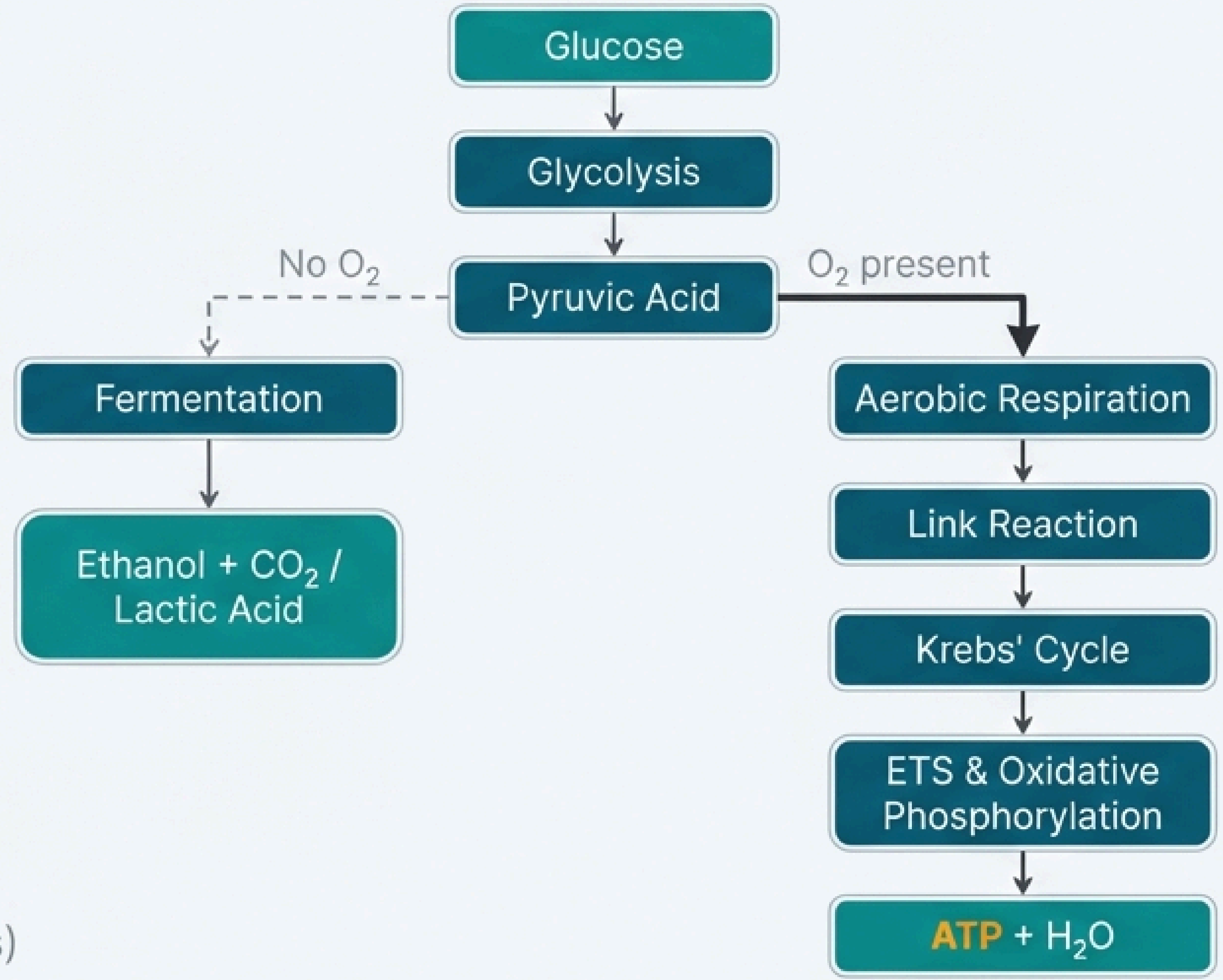
NCERT Syllabus Mapping

- 12.1 Do Plants Breathe?
- 12.2 Glycolysis
- 12.3 Fermentation
- 12.4 Aerobic Respiration
    - 12.4.1 Tricarboxylic Acid Cycle
    - 12.4.2 Electron Transport System (ETS) and Oxidative Phosphorylation
- 12.5 The Respiratory Balance Sheet
- 12.6 Amphibolic Pathway
- 12.7 Respiratory Quotient

NEET Weightage Trend: High

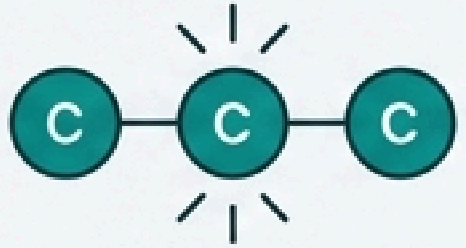
*(Consistently 2-4 questions per year, touching all major pathways)

High-Level Concept Flow



12.1 Do Plants Breathe?

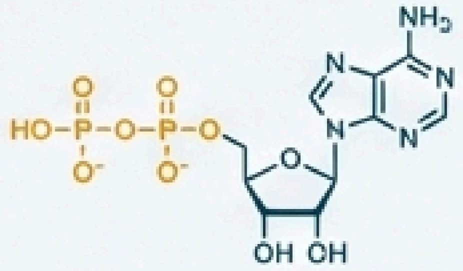
What is Cellular Respiration?



Definition: The breaking of the C-C bonds of complex compounds through oxidation within the cells, leading to the release of a considerable amount of energy.

Respiratory Substrates: Compounds oxidised during this process.
Primary: Carbohydrates (usually glucose)
Secondary: Proteins, fats, and even organic acids under certain conditions.

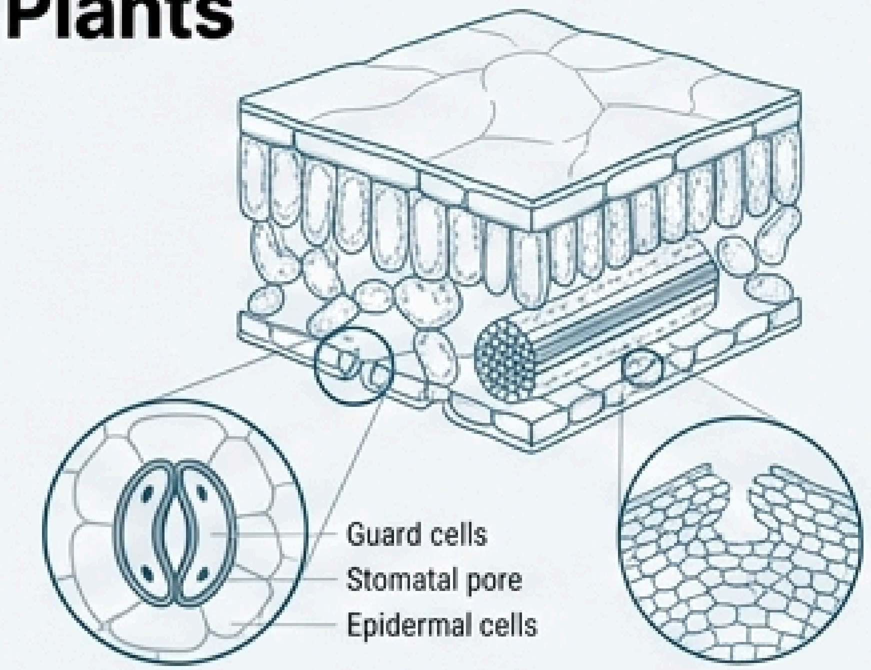
Energy Trapping: The energy is released in a series of slow, step-wise reactions, controlled by enzymes. This energy is trapped in the form of ATP (Adenosine Triphosp), the energy currency of the cell.



Gaseous Exchange in Plants

Plants require O_2 for respiration and release CO_2.

Mechanisms: Gaseous exchange occurs via **stomata** (leaves) and **lenticels** (woody stems).



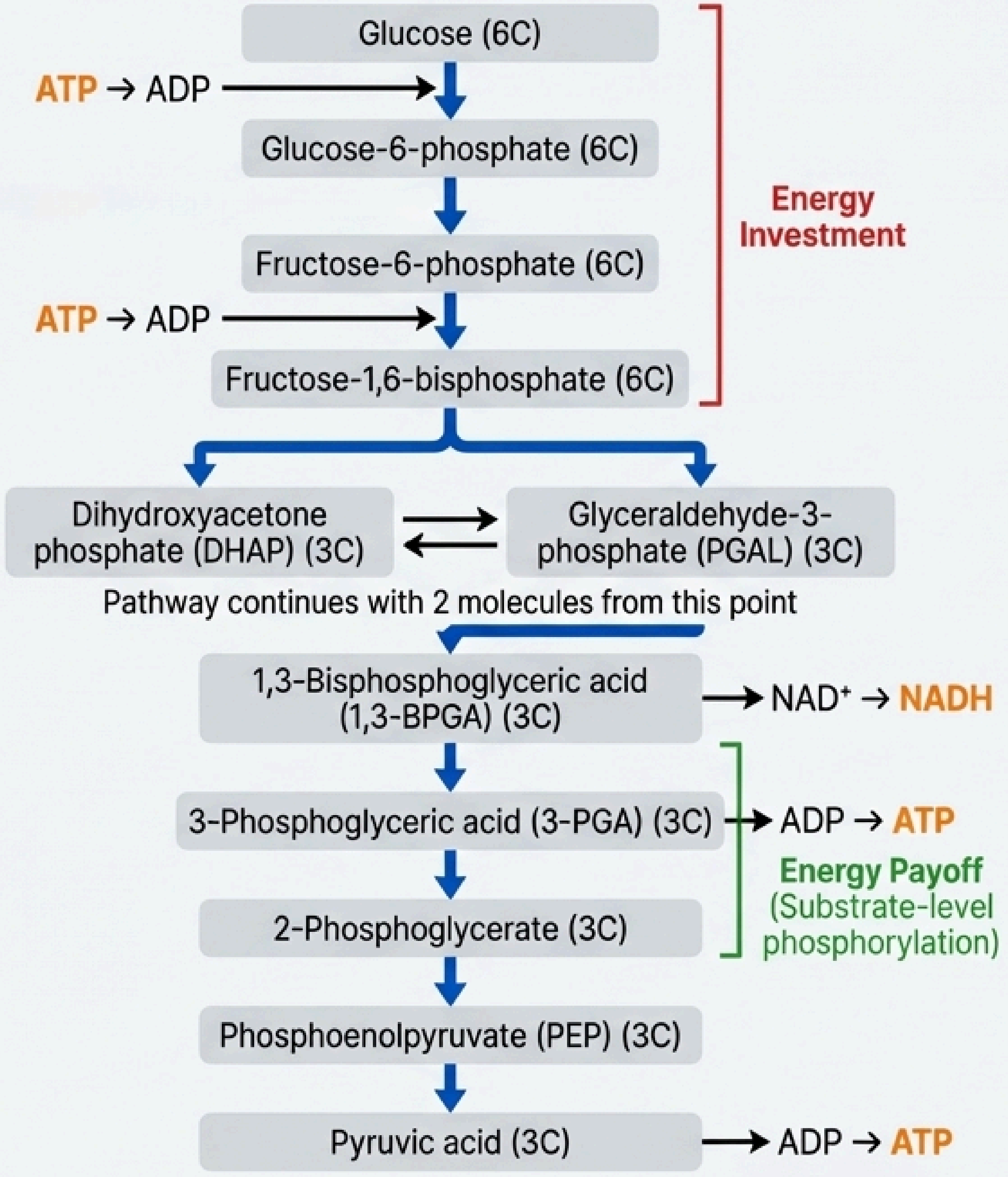
Why plants don't need dedicated respiratory organs:

1. **Localised Needs:** Each plant part takes care of its own gas-exchange needs with little transport between parts.
2. **Lower Demand:** Roots, stems, and leaves respire at rates far lower than animals do.
3. **Short Diffusion Distance:** Most living cells are close to the surface, and loose packing of parenchyma cells creates an interconnected network of air spaces.

12.2 Glycolysis (EMP Pathway)

Location: Cytoplasm

- **Definition:** The breakdown of glucose to pyruvic acid. From Greek *glycos* (sugar) and *lysis* (splitting).
- **Alternative Name:** EMP Pathway, after its discoverers (Embden, Meyerhof, and Parnas).
- **Universality:** Occurs in all living organisms and is the sole process of respiration in anaerobic organisms.
- **Process:** A sequence of 10 enzyme-controlled reactions causing **partial oxidation** of glucose.



Key Steps & Net Products (from 1 Glucose molecule)

- **Input:** 1 Glucose (6C)
- **Output:** 2 Pyruvic Acid (3C)
- **Net Gain:** 2 ATP (4 ATP produced, 2 ATP consumed)
- 2 NADH + $2H^+$
- **Key Product:** Pyruvic acid is the pivotal product connecting glycolysis to subsequent pathways.

12.3 Fermentation

Location: Cytoplasm (under anaerobic conditions)

- **Definition:** The incomplete oxidation of glucose where pyruvic acid is converted to CO_2 and ethanol, or to lactic acid.
- **Purpose:** To regenerate NAD^+ from NADH, allowing glycolysis to continue.

- **Energy Yield:** Very low. Less than 7% of the energy in glucose is released. Net gain is only **2 ATP** (from glycolysis). The process itself is hazardous, producing acid or alcohol.

Types of Fermentation

Feature	Alcoholic Fermentation	Lactic Acid Fermentation
Organisms	Yeast, many prokaryotes 	Some bacteria, animal muscle cells 
Process	Pyruvic acid is first decarboxylated, then reduced to ethanol.	Pyruvic acid is directly reduced to lactic acid.
Enzymes	Pyruvic acid decarboxylase, Alcohol dehydrogenase	Lactate dehydrogenase
Products	**Ethanol** + CO_2	**Lactic Acid**
Hazard	Yeast poisons itself when alcohol concentration reaches about 13%.	Lactic acid accumulation causes muscle fatigue.

12.4 Aerobic Respiration: The Link Reaction

📍 Location: Mitochondrial Matrix

CYTOPLASM

MITOCHONDRIAL MATRIX

- **Prerequisite: Presence of Oxygen (O_2).**
- **Process:** Pyruvate, the final product of glycolysis, is transported from the cytoplasm into the mitochondria for complete oxidation.

Oxidative Decarboxylation (The Link Reaction)

- **Definition:** The reaction that **connects glycolysis to the Krebs' Cycle**, converting **pyruvic acid** into **Acetyl CoA**.
- **Enzyme Complex: Pyruvate Dehydrogenase**
- **Coenzymes Required: NAD$^+$** and **Coenzyme A**

Pyruvic Acid →

$$\underset{(3C)}{\text{Pyruvic Acid}} + \text{CoA} + NAD^+ \xrightarrow{\text{Pyruvate Dehydrogenase}} \underset{(2C)}{\text{Acetyl CoA}} + CO_2 + NADH + H^+$$

Net Products (from 1 Glucose molecule → 2 Pyruvic Acids):

- **2 Acetyl CoA (2C)**
- **2 CO_2**
- **2 NADH + 2H$^+$**

CYTOPLASM

Acetyl CoA is now ready to enter the Tricarboxylic Acid (TCA) Cycle.

12.4.1 Tricarboxylic Acid Cycle (Krebs' Cycle)

Location: Mitochondrial Matrix

Definition: A cyclic pathway where the acetyl group of Acetyl CoA is completely oxidized to CO_2 and water. Named after scientist Hans Krebs.

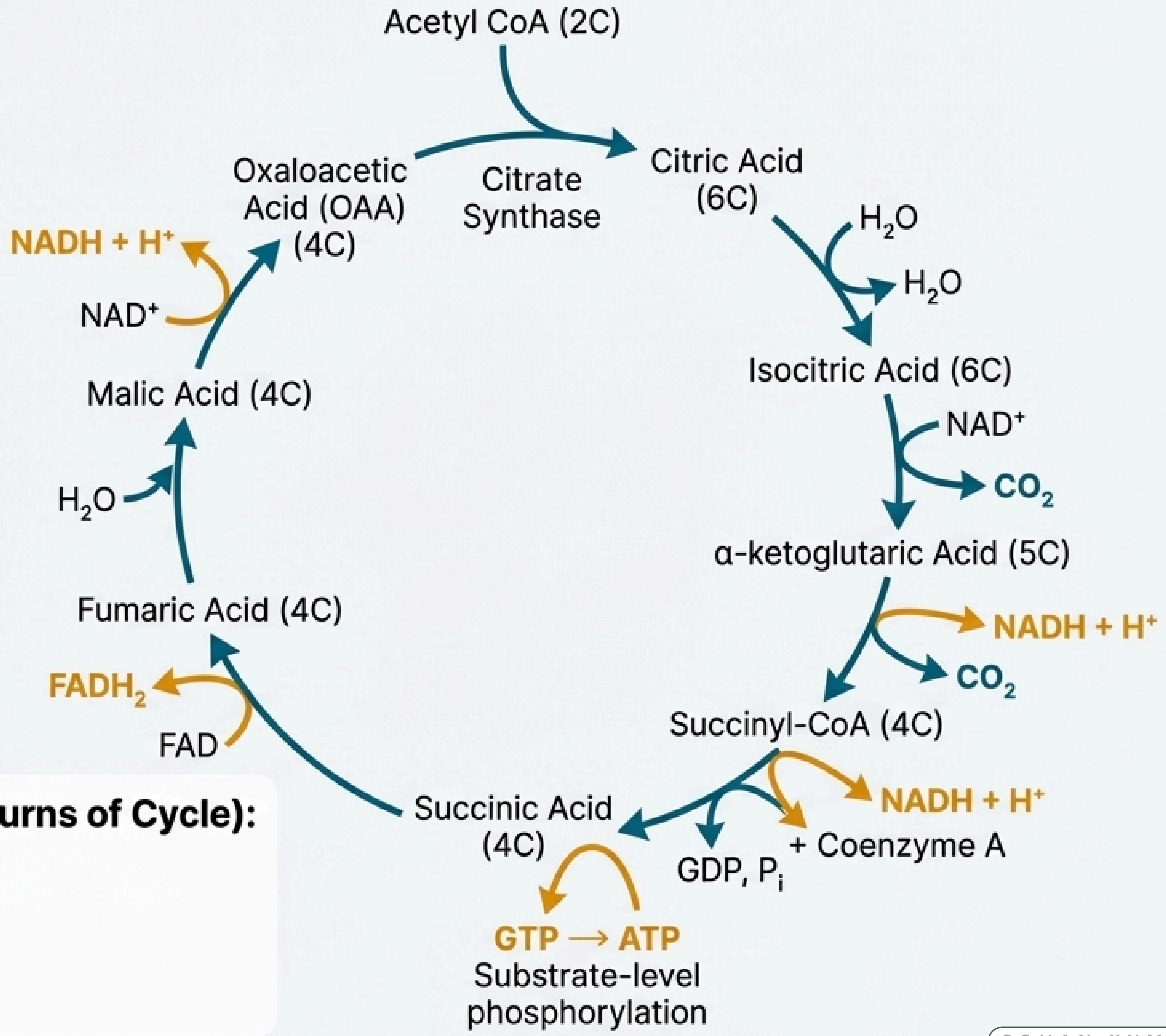
First Step: Condensation of Acetyl CoA (2C) with Oxaloacetic Acid (OAA) (4C) to form Citric Acid (6C), the cycle's first product.

Summary of One Turn (for 1 Acetyl CoA)

- **CO_2 Released:** 2 molecules
- **NADH + H^+ Produced:** 3 molecules
- **$FADH_2$ Produced:** 1 molecule
- **ATP/GTP Produced:** 1 molecule

Net Products (from 1 Glucose → 2 Acetyl CoA → 2 Turns of Cycle):

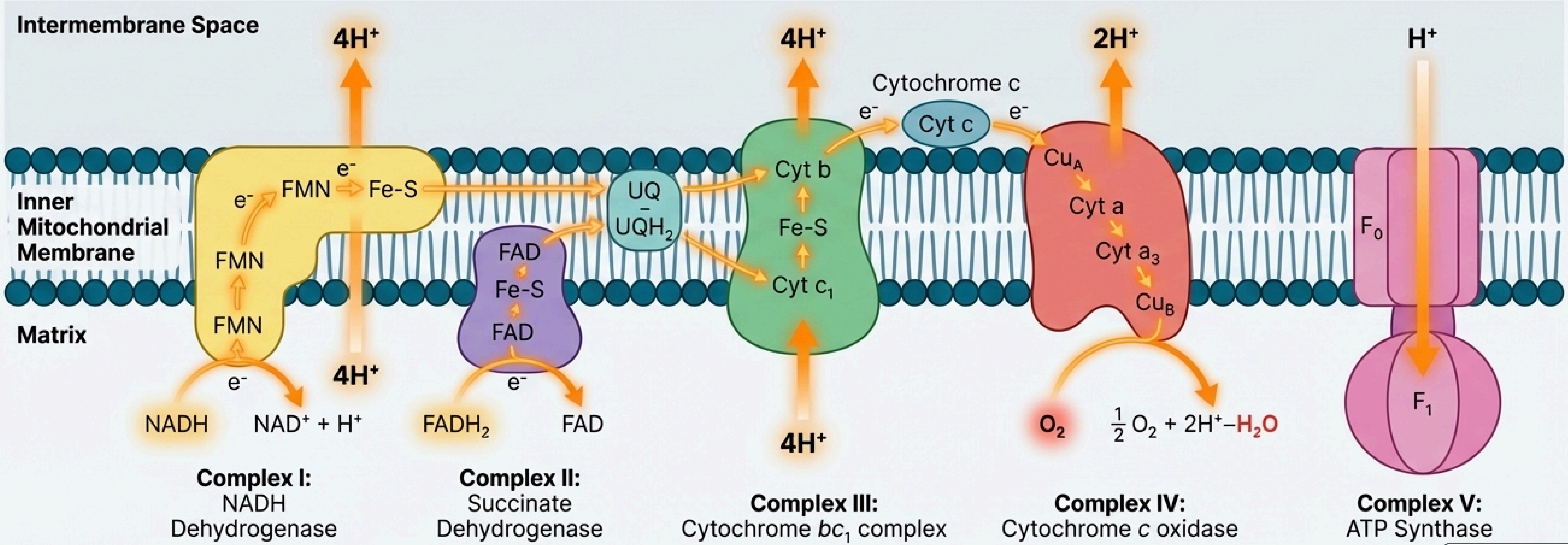
- 4 CO_2
- 6 NADH + 6H^+
- 2 $FADH_2$
- 2 ATP/GTP



12.4.2 Electron Transport System (ETS)

📍 Location: Inner Mitochondrial Membrane

- **Purpose:** To release and utilise the energy stored in NADH and $FADH_2$ (produced during glycolysis and Krebs' cycle).
- **Process:** Electrons are passed down a series of protein carriers, releasing energy in steps. This energy is used to pump protons (H^+) from the matrix into the intermembrane space, creating a proton gradient.
- **Terminal Electron Acceptor: Oxygen (O_2) is vital**, as it is the final acceptor of electrons and protons, forming water. This drives the entire process.



12.4.2 Oxidative Phosphorylation

📍 Location: Inner Mitochondrial Membrane (using Complex V)

Definition:

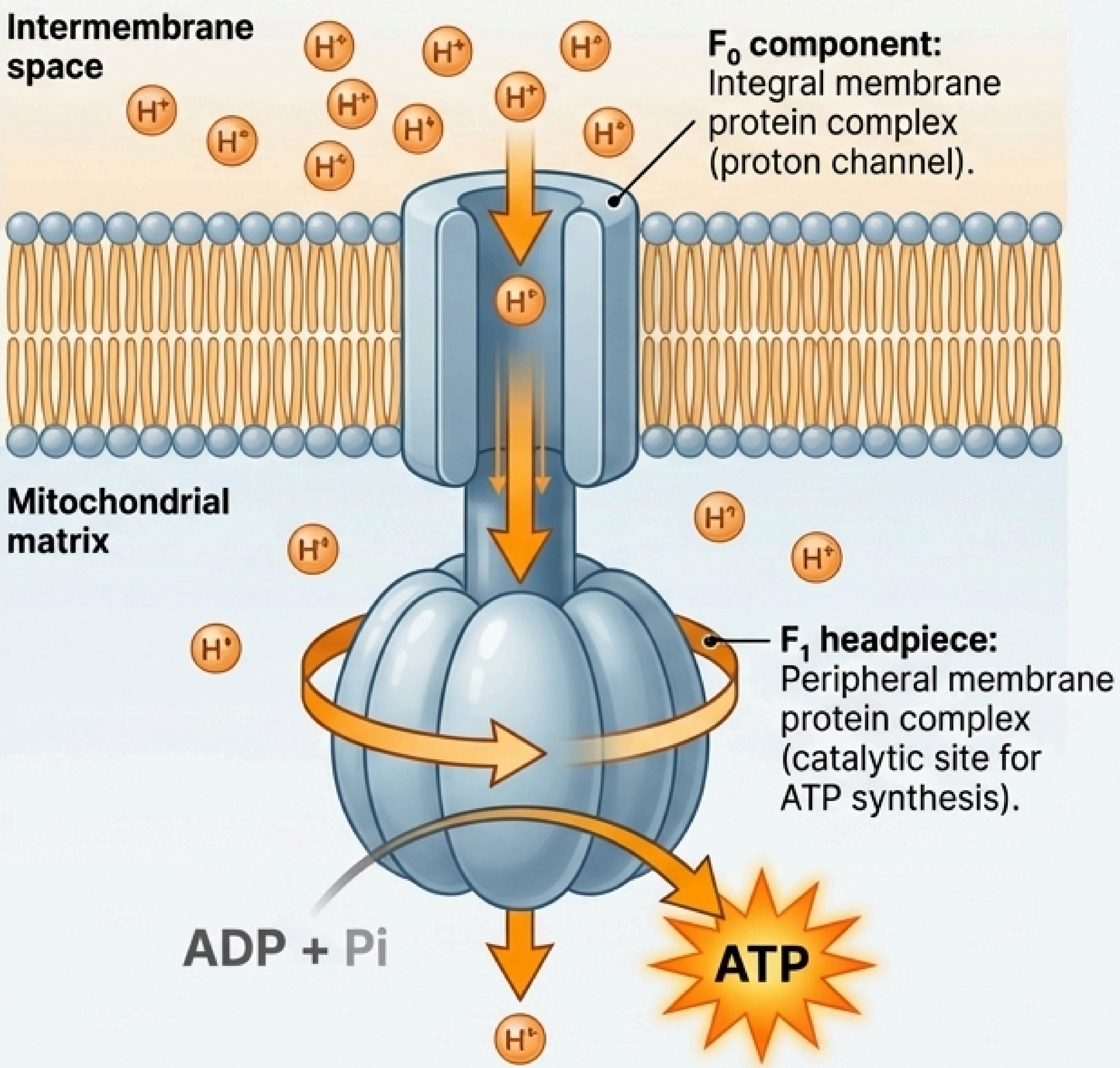
The synthesis of ATP using energy released by the oxidation-reduction reactions in the ETS. This is why it's called oxidative phosphorylation.

Mechanism (Chemiosmotic Hypothesis):

1. The proton gradient (high H^+ in intermembrane space) created by the ETS acts as a form of stored energy.
2. Protons flow back into the matrix down their electrochemical gradient.
3. This flow occurs through a specific channel in Complex V (ATP Synthase).
4. The energy of this proton flow drives the synthesis of ATP from ADP and inorganic phosphate (Pi).

Energy Yield:

Oxidation of one molecule of **NADH** yields **3** molecules of **ATP**.
Oxidation of one molecule of **$FADH_2$** yields **2** molecules of **ATP**.



12.5 The Respiratory Balance Sheet & 12.6 Amphibolic Pathway

38 ATP

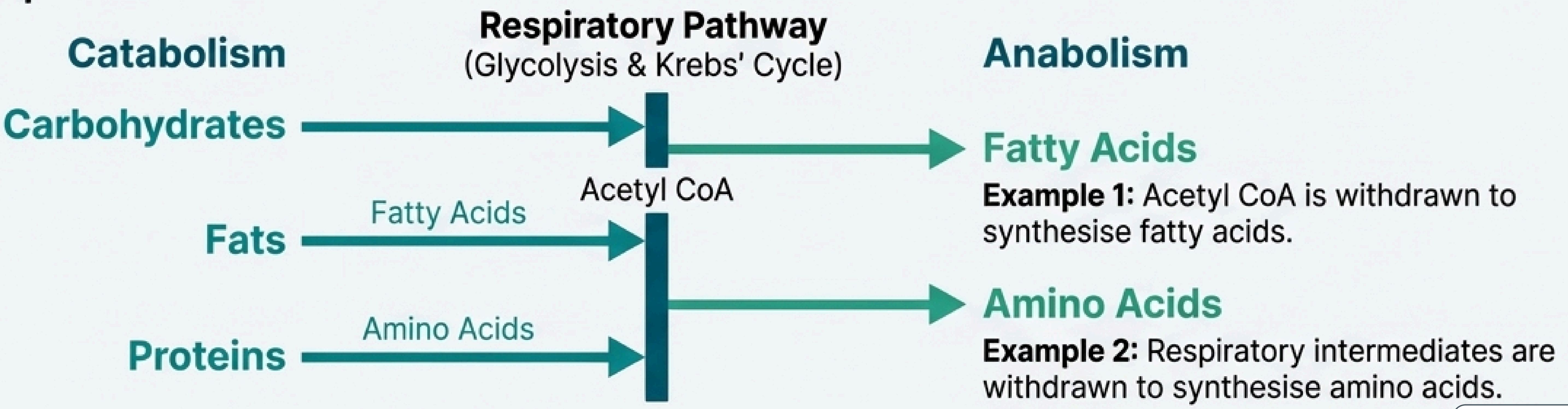
Theoretical Net Gain per glucose molecule.

- **This is a theoretical calculation based on several assumptions:**
  1. The pathway is sequential and orderly (Glycolysis → TCA → ETS).
  2. NADH from glycolysis is efficiently transported into the mitochondria.
  3. No pathway intermediates are withdrawn for synthesis of other compounds.
  4. Only glucose is the substrate.

In a living system, these assumptions are not strictly valid, as pathways work simultaneously and intermediates are often diverted.

Definition: A pathway involved in both **breakdown (catabolism) and synthesis (anabolism).**

Respiration is Amphibolic:

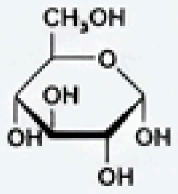
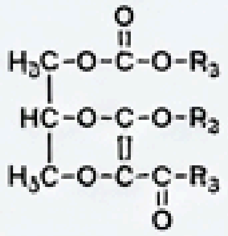
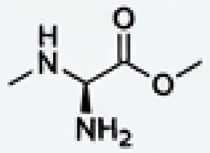


12.7 Respiratory Quotient (RQ)

The ratio of the volume of CO_2 evolved to the volume of O_2 consumed during respiration.

$$RQ = \frac{\text{Volume of } CO_2 \text{ evolved}}{\text{Volume of } O_2 \text{ consumed}}$$

The RQ value depends on the type of respiratory substrate being used.

RQ Values for Different Substrates		
Substrate	**RQ Value**	**Equation & Explanation**
Carbohydrates 	**1.0**	$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ $RQ = 6\ CO_2 / 6\ O_2 = 1.0$ Equal volumes of CO_2 and O_2 are evolved and consumed.
Fats 	**< 1** (~0.7 for Tripalmitin)	$2(C_{51}H_{98}O_6) + 145O_2 \rightarrow 102CO_2 + 98H_2O$ $RQ = 102\ CO_2 / 145\ O_2 \approx 0.7$ Fats are poorer in oxygen and require more O_2 for complete oxidation.
Proteins 	**~ 0.9**	Proteins are also less rich in oxygen compared to carbohydrates.

In living organisms, respiratory substrates are often a mix, so pure RQ values are rare.

NCERT Line Emphasis

⚠ **NCERT Line:** *'The breaking of the C-C bonds of complex compounds through oxidation within the cells, leading to release of considerable amount of energy is called respiration.'*
★ **Direct NEET Pickup:** The core definition of cellular respiration.

⚠ **NCERT Line:** *'Pyruvic acid is then the key product of glycolysis. What is the metabolic fate of pyruvate? This depends on the cellular need.'*
★ **Direct NEET Pickup:** Establishes pyruvate as the central connecting molecule.

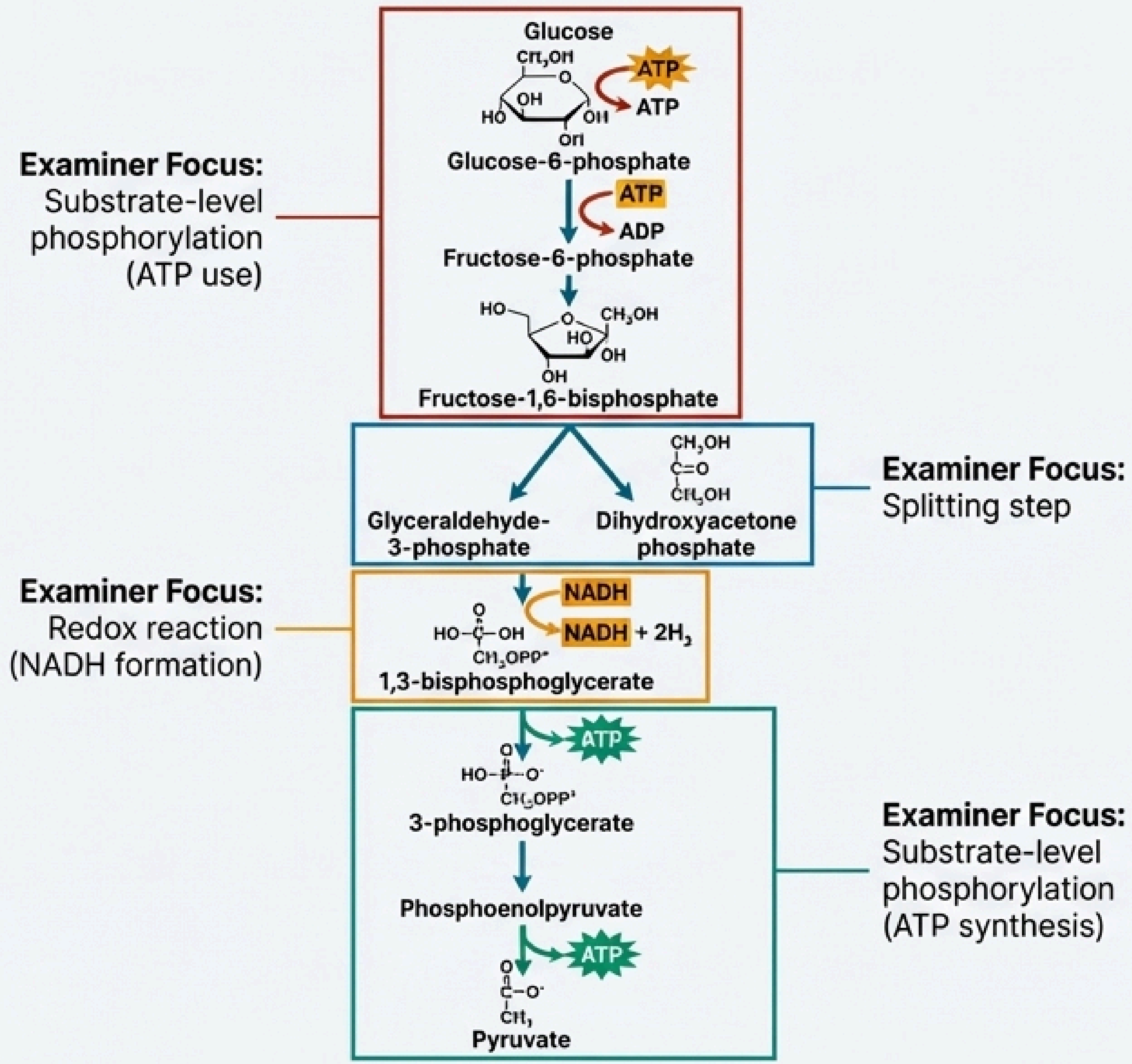
⚠ **NCERT Line:** *'In both lactic acid and alcohol fermentation not much energy is released; less than seven per cent of the energy in glucose is released...'*
★ **Direct NEET Pickup:** A specific value frequently asked to highlight the inefficiency of fermentation.

⚠ **NCERT Line:** *'Oxygen acts as the final hydrogen acceptor.'*
★ **Direct NEET Pickup:** The precise role of oxygen in the ETS is a critical and common question.

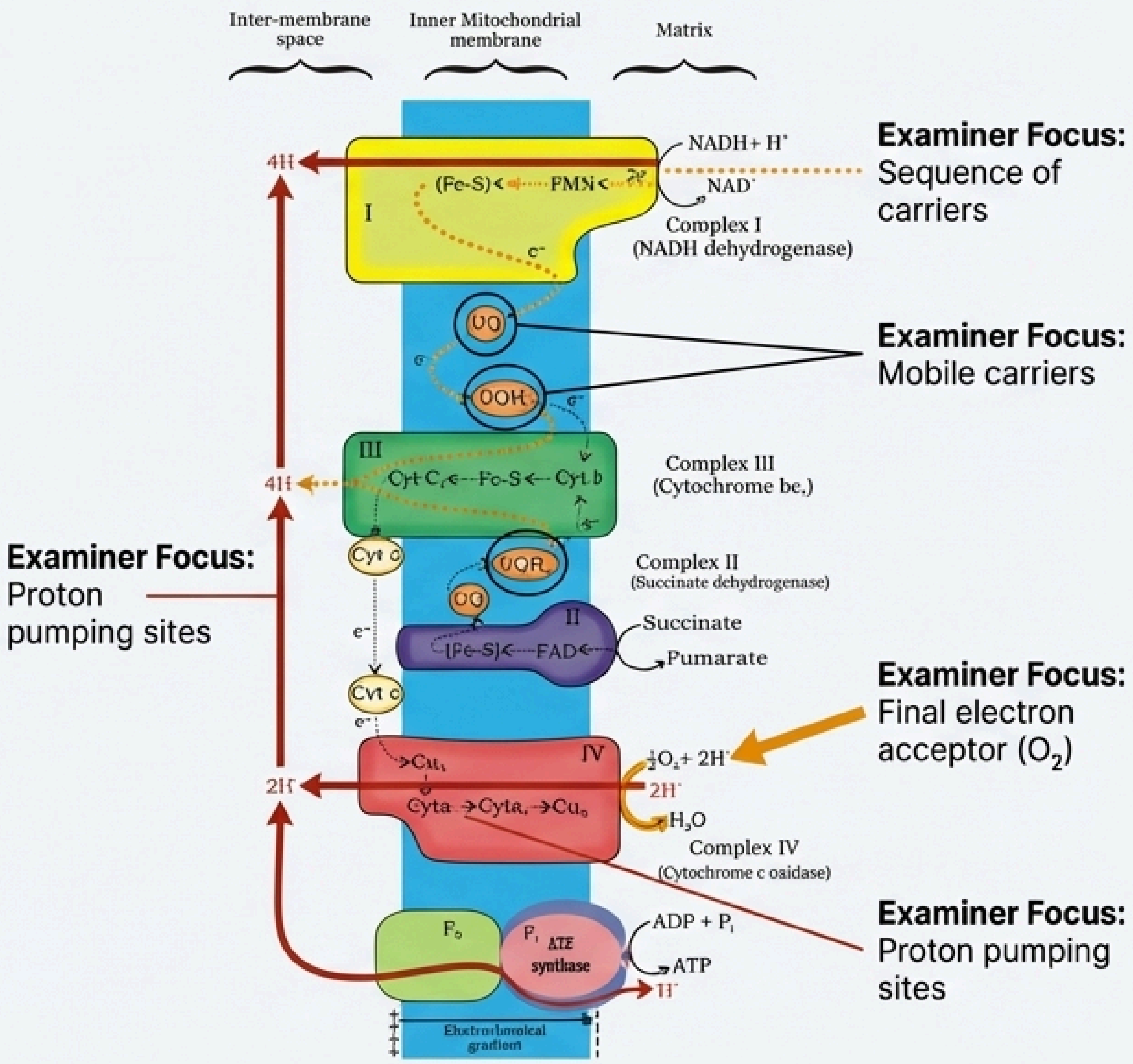
⚠ **NCERT Line:** *'Because the respiratory pathway is involved in both anabolism and catabolism, it would hence be better to consider the respiratory pathway as an amphibolic pathway rather than as a catabolic one.'*
★ **Direct NEET Pickup:** This justifies the term 'amphibolic' and is a hotspot for conceptual questions.

Diagrams & Visual Explanation

1. Glycolysis Pathway (Figure 12.1)



2. Electron Transport System (ETS) (Figure 12.4)



Comparison & Difference Tables

Glycolysis vs. Krebs' Cycle

Parameter	Glycolysis	Krebs' Cycle
Location	**Cytoplasm**	**Mitochondrial Matrix**
Nature of Pathway	Linear	Cyclic
Starting Substrate	Glucose (6C)	Acetyl CoA (2C)
Oxygen Requirement	Anaerobic (O_2 not directly required)	Strictly Aerobic (requires O_2 to regenerate NAD^+/FAD)
Key Products (per glucose)	2 Pyruvic Acid, **2 ATP** (Net), 2 NADH	4 CO_2, **2 ATP/GTP**, 6 NADH, 2 $FADH_2$

Aerobic Respiration vs. Fermentation

Parameter	Aerobic Respiration	Fermentation
Oxygen	Required	Not Required
Oxidation of Glucose	Complete (to CO_2 + H_2O)	Incomplete (to Alcohol or Lactic Acid)
Net ATP Gain	High (Theoretically **38 ATP**)	Low (Only **2 ATP**)
Location	**Cytoplasm & Mitochondria**	**Cytoplasm only**
NADH Oxidation	Vigorous (in ETS)	Slow (to regenerate NAD^+)
Final e^- Acceptor	O_2	Pyruvic acid or its derivatives

NEET-UG Focus & Assertion-Reason

NEET-UG Focus

Previous Year Themes:

- Location of each pathway (Matrix matching).
- Net gain of ATP/NADH/$FADH_2$ from each stage.
- The role of Oxygen as the final electron acceptor in ETS.
- RQ values for carbohydrates, fats, and proteins.
- Connecting link molecules: Pyruvic Acid and Acetyl CoA.

Common Traps & Misconceptions:

💥 **Trap:** Confusing NET gain of ATP (2) with GROSS production (4) in glycolysis.

💥 **Misconception:** Thinking O_2 is directly used in the Krebs' cycle. (It's needed for the ETS to regenerate NAD^+ and FAD).

💥 **Trap:** Forgetting that Krebs' cycle calculations must be doubled for one glucose molecule.

Memory Hook (Krebs' Cycle Intermediates):

Citrate Is Krebs' Starting Substrate For Making Oxaloacetate.

Assertion–Reason & Statement Analysis

Assertion (A): The respiratory pathway is considered an amphibolic pathway.

Reason (R): It involves both breakdown (catabolism) and synthesis (anabolism) of substrates.

Analysis: Both A and R are true, and R is the correct explanation of A.

Statement 1: Fermentation accounts for the complete breakdown of glucose to CO_2 and H_2O.

Statement 2: In fermentation, NADH is oxidised to NAD^+ rather slowly.

Analysis: Statement 1 is False (it's partial breakdown).

Statement 2 is True.

Assertion (A): The RQ of fats is less than 1.

Reason (R): Fats require more O_2 for their complete oxidation compared to the amount of CO_2 evolved.

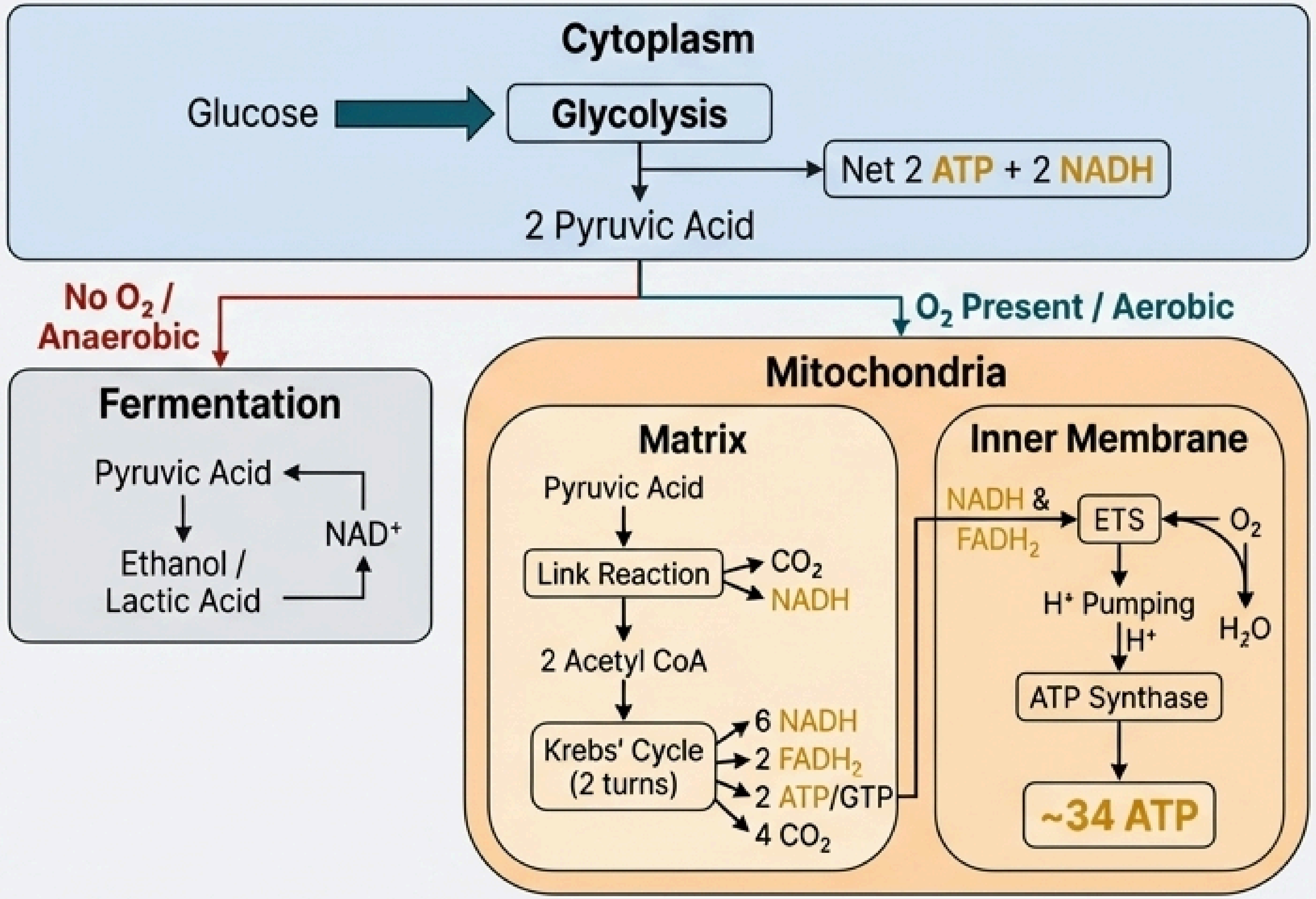
Analysis: Both A and R are true, and R is the correct explanation for A.

Rapid Revision & Final Chapter Soter Snapshot

Rapid Revision: Bullet Points

- **Glycolysis:** Cytoplasm, 1 Glucose → 2 Pyruvate, Net 2 ATP, 2 NADH.
- **Fermentation:** Anaerobic, Cytoplasm, regenerates NAD^+, Net 2 ATP yield.
- **Link Reaction:** Mitochondrial Matrix, Pyruvate → Acetyl CoA + CO_2 + NADH.
- **Krebs' Cycle:** Mitochondrial Matrix. Per Glucose: 4 CO_2, 6 NADH, 2 $FADH_2$, 2 GTP.
- **ETS:** Inner Mitochondrial Membrane, uses O_2 as final e^- acceptor, creates a proton gradient.
- **Oxidative Phosphorylation:** Uses proton gradient via ATP Synthase to synthesize the bulk of ATP.
- **Yield**: 1 NADH → 3 ATP; 1 $FADH_2$ → 2 ATP.
- **Amphibolic**: Pathway is used for both breakdown and synthesis.
- **RQ:** Carbohydrates = 1.0, Fats < 1.0, Proteins ≈ 0.9.

Final Chapter Snapshot: The Journey of a Glucose Molecule



If you remember only this much: Glucose breaks down universally in the cytoplasm (Glycolysis). With oxygen, its products enter the mitochondria for a massive ATP yield via the Krebs' Cycle and ETS. Without oxygen, it undergoes fermentation for a small but crucial ATP yield.